

CSE 220 Online on DFT
Section A

Time: 30 minutes

Scenario:

You are given two large, non-negative integers represented as arrays of their **base-10 digits**. For computational convenience, the arrays are ordered from the **Least Significant Digit (LSD)** to the **Most Significant Digit (MSD)**.

Let the two numbers be A and B , with lengths m and n respectively:

$$A = [a_0, a_1, \dots, a_{m-1}] \quad \text{representing} \quad \sum_{i=0}^{m-1} a_i \cdot 10^i$$

$$B = [b_0, b_1, \dots, b_{n-1}] \quad \text{representing} \quad \sum_{j=0}^{n-1} b_j \cdot 10^j$$

Task:

Compute the **product** of these two large integers. You must output the result as an array of digits (also ordered LSD to MSD) using **FFT/IFFT-based circular convolution**.

Hints:

- Multiplying two numbers in base-10 is mathematically equivalent to multiplying two polynomials evaluated at $x = 10$.
- Computing the coefficients $C[k] = \sum_{i+j=k} a_i \cdot b_j$ is exactly the same as performing a **linear convolution** of the digit arrays.
- Linear convolution can be computed directly, **but in this online, you will have to implement this using circular convolution via FFT/IFFT**.
- Remember to pad the sequences with zeros to a suitable length (at least $m + n - 1$) to prevent circular wrap-around (aliasing).
- **Carry Propagation:** After convolution, the resulting array elements might be ≥ 10 . You must iterate through your resulting array, keeping $C[k] \pmod{10}$ at the current position, and adding the carry $\lfloor C[k]/10 \rfloor$ to the next position $C[k+1]$.

Example Input 1:

Multiply 123 by 45. Represented as LSD-first arrays:

$$A = [3, 2, 1]$$

$$B = [5, 4]$$

Expected Output 1:

Result: $[5, 3, 5, 5]$ (Which represents 5535)

Calculation Steps:

- **Step 1: Convolution**

Compute $C[k] = \sum a_i b_{k-i}$ for the padded arrays:

$$C[0] = a_0 \cdot b_0 = 3 \cdot 5 = 15$$

$$C[1] = a_0 \cdot b_1 + a_1 \cdot b_0 = 3 \cdot 4 + 2 \cdot 5 = 12 + 10 = 22$$

$$C[2] = a_1 \cdot b_1 + a_2 \cdot b_0 = 2 \cdot 4 + 1 \cdot 5 = 8 + 5 = 13$$

$$C[3] = a_2 \cdot b_1 = 1 \cdot 4 = 4$$

Raw Convolution Array: $[15, 22, 13, 4]$

- **Step 2: Carry Propagation**

Index 0: Keep $15 \pmod{10} = 5$. Carry $\lfloor 15/10 \rfloor = 1$.

Index 1: $22 + 1(\text{carry}) = 23$. Keep 3. Carry 2.

Index 2: $13 + 2(\text{carry}) = 15$. Keep 5. Carry 1.

Index 3: $4 + 1(\text{carry}) = 5$. Keep 5. Carry 0.

Final Normalized Array: $[5, 3, 5, 5]$

Example Input 2:

Multiply 999 by 99.

$$A = [9, 9, 9]$$

$$B = [9, 9]$$

Expected Output 2:

Result: $[1, 0, 9, 8, 9]$ (Which represents 98901)

Note: The raw convolution will be $[81, 162, 162, 81]$. You must correctly propagate the carries all the way through.

Submission

Just submit a single python file after renaming it with your student ID: 2205XXX.py